

The Synthetic-to-Real Gap in Sheet-Image Score Following is Representational, not Acoustic

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ABSTRACT

Score followers that track a live performance on a sheet-music image degrade sharply when moved from synthesised audio to real recordings. This degradation is routinely attributed to acoustic mismatch — reverberation, microphone response, instrument timbre — and is addressed with acoustic remedies. We show that this attribution is largely wrong. Using two microphones placed on the *same* performances, so that the performance is held exactly fixed, we measure that room reverberation costs a stock automatic music transcription (AMT) system essentially nothing: onset F_1 is 0.9116 on the room microphone versus 0.9124 on a direct pickup at 50 ms tolerance. On those same recordings, image-conditioned trackers lose roughly thirty points of tracking accuracy. The note-level information survives the room almost intact; the tracking architecture discards it. We corroborate this with an intervention asymmetry: augmentation with real measured impulse responses, drawn from one bank and applied under one protocol, is reported to be worth +25.2 points to a detection-based tracker but is worth only +11.0 to a dense-heatmap tracker. We argue the binding constraint is the output parameterisation rather than the audio front end, and we report three measurement problems — an aggregation mismatch worth +7.3 points, an effective sample size of 25 rather than 4415, and undisclosed oracle inputs — that make small reported gains on this benchmark uninterpretable.

Index Terms— score following, music alignment, optical music recognition, evaluation, domain shift

1 Introduction

Sheet-image score following takes a live audio performance and a rendered image of the score, and predicts the performer’s current position on the image, frame by frame. Unlike symbolic score following, it consumes the score as pixels, which removes any need for a machine-readable encoding of the piece.

Systems in this family are trained almost exclusively on synthesised audio, because ground truth requires an exact alignment between audio and notehead coordinates, and that is cheap to obtain only by rendering. Every such system degrades when evaluated on real recordings. The degradation is large: on the real-audio subset of MSMD, published CUNet scores 22.4 where the same architecture reaches the high eighties on synthetic audio.

The prevailing explanation is acoustic. Real rooms add reverberation, real microphones colour the spectrum, and real pianos differ from the soundfont, so the audio encoder sees inputs unlike anything in training. This explanation motivates acoustic remedies — impulse-response augmentation, multi-condition training, test-time normalisation — and it is the

framing under which we ourselves worked for several months.

This paper argues that the explanation is mostly incorrect, and that acting on it has misdirected effort.

Our contributions:

1. A controlled measurement showing that the room preserves note-level information almost perfectly (Section 3). Two microphones on the same take isolate the room with the performance held fixed.
2. An intervention asymmetry showing that the same acoustic remedy pays very differently depending on the output parameterisation (Section 4).
3. Three measurement problems on this benchmark that jointly make gains below roughly ten points uninterpretable (Section 5), with a corrected reporting protocol.

2 Setup

Data. We use MSMD and its real-recording extension. We evaluate on four acoustic tiers over the same 25 pages: *synth* (soundfont rendering, matching training); *rp_synth* (a matched synthetic control: same pages, same ground truth, training soundfont); *di-left* (real piano, direct pickup, no room); and *room* (real piano, room microphone). Because *di-left* and *room* are two microphones recording the *same* take, their difference isolates the room with the performance held exactly fixed. This pairing is the instrument used throughout Section 3.

Metric. We report PCT@0.5s: the percentage of onsets whose predicted position, mapped back through the score-to-onset interpolation, falls within 0.5 s of the true onset. Thresholds are {0.05, 0.1, 0.5, 1.0, 5.0} s. All numbers are onset-only and *micro*-aggregated (onsets pooled across pieces, then divided), matching the convention of the baselines we compare against. Section 5 shows why that choice must be stated.

Baselines. We reproduced the strongest reproducible published system ourselves rather than quoting it, obtaining 63.0 at 0.1 s and 79.9 at 0.5 s on *room*, which matches the published row exactly. Numbers frequently quoted as 63.0 for this system are the 0.1 s column; the 0.5 s column is 79.9.

3 Does the room destroy the information?

If the synthetic-to-real gap were acoustic in the strong sense — the room degrading the audio so that note events are no longer recoverable — then any system reading notes from that audio should degrade too. We test this directly with an

Table 1: Note-onset F_1 on the same 25 performances, two microphones, same take. The room costs a transcriber essentially nothing.

model	tol	room	di-left	Δ
KONG	50 ms	0.9116	0.9124	-0.001
KONG	100 ms	0.9546	0.9903	-0.036
EDWARDS	50 ms	0.9441	0.9259	+0.018
EDWARDS	100 ms	0.9864	0.9909	-0.005

off-the-shelf AMT system, which shares no components with our tracker and was trained on entirely different data.

We transcribe all 25 real performances in both microphone conditions with two AMT models: a widely used piano transcription model (KONG) and a reverb-augmented variant (EDWARDS). We score note onsets with `mir_eval` at 50 and 100 ms.

Table 1 is the central measurement of this paper. At 50 ms the room costs KONG 0.001 F_1 , and the reverb-augmented model is in fact *better* on the room microphone than on the direct pickup. At 100 ms the largest penalty is 0.036. On these same files, our trackers lose roughly thirty points of PCT@0.5s.

Validating the measurement path. Before trusting these numbers we verified that the coordinate round-trip is lossless. Feeding the ground-truth MIDI in as if it were a transcription, and decoding position through the same score-note-to-pixel-to-onset-frame path used for the AMT output, yields exactly 100.00 PCT@0.5s when the tracker is allowed to hear the note it is scored on, and 94.79 under the stricter convention where it is not. A coordinate or indexing error anywhere in this path would show up as a ceiling below 100.

What this does and does not show. It shows the note information survives the room. It does *not* show that a decomposed transcribe-then-align system is a better score follower. Offline dynamic time warping over these transcriptions reaches 98.06 on *room*, but that figure is not comparable to an online tracker: it is non-causal, and it is close to tautological on this data, because the recordings are reproducing-piano playback of the score MIDI and therefore contain no tempo deviation for the alignment to resolve. The corresponding *online* greedy matcher reaches only 10.7, with a median error of eight to thirteen seconds; it loses the pointer and never recovers. Causal matching, not transcription, is the hard part of that pipeline.

4 Where the gap actually is

4.1 The same remedy pays differently

If the audio is not the problem, acoustic augmentation should help less than expected — and should help *unevenly* across architectures. It does.

Augmenting with real measured impulse responses is reported to move a detection-based tracker from 46.0 to 71.2 on *room*, a gain of +25.2. We applied augmentation from a

bank of 693 real measured impulse responses to our own dense-heatmap tracker, re-encoding degraded waveforms through the frozen audio front end so the degradation cannot be undone by a normalisation layer. The gain was +11.0 (45.6 $\rightarrow$ 56.6): real, the largest single improvement we have obtained, and well under half of what the same class of intervention buys a detection model.

A negative result we must disclose. Our own earlier conclusion that impulse-response augmentation does not help was an artefact. The convolution used `mode='same'`, which advances the signal by $(|h| - 1)/2$ samples relative to its onset labels — a 4 to 20 frame desynchronisation on half of all augmented samples. The experiment that produced the +11.0 above differs from the discarded one chiefly in that this bug is fixed and the impulse responses are measured rather than synthesised. We report this because the failure mode is silent: the training loss is unremarkable and only the real-audio metric suffers.

4.2 The output parameterisation

A controlled comparison in the literature points to the same place. Within one paper, on one dataset, with the same audio front end, a system that predicts position as a *bucketed softmax classification* scores 58.5 on *room* where the dense soft-Dice heatmap formulation scores 22.4 — a 36-point difference attributable to the output layer alone. This is consistent with the segmentation literature, where soft-Dice is known to perform well in-distribution and poorly out-of-distribution, and synthetic-to-real is exactly such a shift.

Three independent observations therefore converge: the audio retains the information (Section 3); acoustic remedies transfer only partly to heatmap models; and the largest controlled swing in the literature is attributable to the output parameterisation. We take this as the strongest available guidance for where effort in this task should go.

5 Measurement problems on this benchmark

Alongside the above we found three properties of the standard evaluation that affect how any result on it should be read.

Aggregation. Reported numbers in this literature use two different estimators without stating which. Pooling all onsets and dividing (*micro*) and averaging per-piece percentages (*macro*) are not interchangeable: on one unchanged checkpoint we measure 45.4 micro, 46.8 macro over pages, and 52.7 macro over pieces. The spread of +7.3 points arises because a single difficult nocturne constitutes 29.8% of the micro metric but only 6.2% of the piece-macro metric, and it is the recording on which every system we have tested performs worst. Macro aggregation systematically discounts the hardest part of the benchmark.

Effective sample size. Onsets within a piece are not independent. A piece-level cluster bootstrap gives a design effect of approximately 180, so the effective sample size of the real-audio benchmark is about 25 pieces, not 4415 onsets. The minimum detectable difference is roughly ten points. We confirmed the practical consequence directly: retraining one

recipe twice, changing nothing but the random seed and the run, produced 22.7 and 16.4 — a spread larger than many published ablation effects.

Undisclosed oracle inputs. Systems differ in what structure they are given versus what they must infer. In one recent system, ground-truth system and bar bounding boxes are required arguments of the forward pass at both training and inference; its loss contains no localisation term of any kind. It selects among given regions, whereas the baseline it is compared against must produce those regions from the image. This is not stated in the paper and is visible only in the released code.

Recommended protocol. We suggest that work on this benchmark report (i) both micro and macro aggregation, (ii) piece-level bootstrap confidence intervals rather than point estimates, (iii) results across acoustic tiers rather than a single condition, including a matched synthetic control, and (iv) an explicit statement of which structures the model receives as input and which it must predict.

6 Conclusion

The synthetic-to-real gap in sheet-image score following is not primarily an acoustic problem. A transcriber recovers 91–95% of onsets from the same room recordings on which trackers lose thirty points, and loses essentially nothing to the room when the performance is held fixed. The information is present in the audio and is discarded downstream. Consistent with this, acoustic augmentation transfers only partially to dense-heatmap models, and the largest controlled effect in the literature is attributable to the output parameterisation. We also find that the benchmark, as commonly used, cannot support the size of the improvements typically claimed on it.